

Eye-Blink artifact Detection and Removal Approaches for BCI using EEG

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Abstract— Electroencephalography (EEG) has been widely used in the development of a BCI system that can translate and decode user goals without clinical procedures. Unfortunately, because of volume conduction, the spatial resolution of the multichannel electroencephalogram (EEG) is poor. The signals from various areas of the head are recorded by each EEG electrode. There are various artifacts like ECG, EMG, and eye movement artifacts present in the EEG. MI-based Brain-Computer Interface (BCI) helps motor impaired persons to connect to the outside world by carrying out a set of motor functions. The BCI system commonly includes pre-processing of raw brain signals, extraction of significant features and classification. Pre-processing plays an important role in the performance of the BCI system. In this paper, three pre-processing methods namely Band pass filtering, Common average referencing and Surface Laplacian are effectively implemented to reduce the various artifacts present in EEG signal thereby improving the performance of the BCI system.

Keywords— Brain-computer interface (BCI), Electroencephalography (EEG), Motor imagery (MI), spatial resolution.

I. INTRODUCTION

The basic goal of a motor imagery brain-computer interface (BCI) is to categorize motor intentions by extracting motor imagery-related properties from electroencephalogram (EEG) signals[1]. Electroencephalography (EEG) has been widely used to construct a BCI system that can translate and decode the goals of users without doing invasive clinical practices.

Thus, the EEG can be used in day-to-day life due to its non-invasiveness, ease of use, and low cost. MI-based BCI can benefit patients with spinal cord impairment and lock-in syndrome by recognizing motor tasks without affecting the spinal cord or the peripheral nervous system.[1]. The BCI system includes preprocessing of raw EEG signals coming from different electrodes attached on brain lobes, extracting significant features from the filtered EEG signals and finally classification of the movements imagined by the subject. Preprocessing plays a vital role in the analysis of BCI system. Due to volume conduction, the spatial resolution of the multichannel electroencephalogram (EEG) is limited. The spatial resolution of an electrode refers to how closely the EEG at that electrode "matches" the activity generated below it in the brain. The signals from various areas of the head are recorded by each EEG electrode. Scalp electrodes are commonly used in humans to measure electrical brain activity, which is electrical events produced several centimeters below the recording electrodes. Consequently, the reported behavior is a mixture (weighted sum) of the

underlying brain sources at each spatial scalp location. This volume-driven mixture is the major cause of the low spatial resolution of EEG (around 5 to 9 cm).[4]

The EEG signal is made up of several different components that originate from various areas of the brain. These components' amplitudes are frequently similar. Therefore, they are not easily identifiable. Blinking in the frontal-polar electrodes (Fp1, Fp2) creates a wave of large amplitude. The amplitude of the wave decreases when the distance from the frontal electrodes is increased. However, it can still be seen in many electrodes. The blinking wave is not of scientific origin; it is thus considered an artifact.[3]. Due to volume conduction, the EEG contains multiple artifacts such as ECG, EMG, and eye movement artifacts, which are caused by a mixing of signals coming from different areas of the brain. Due to the various artifacts present in the signal, it is very difficult to extract the important features necessary for decoding the movements in the motor imagery BCI.

In a review of 96 BCI investigations, 32% used surface Laplacian for spatial filtering, 22% employed principal component or independent component analysis, 14% used common spatial patterns, and 11% used common average reference.[5]. Hence in this paper, three efficient pre-processing methods namely Band-pass filtering, Common Average Referencing and Surface Laplacian (Small and Large surface Laplacian) are implemented to reduce the artifacts present in the signal to improve the performance of the BCI system.

The following is the structure of this paper: The dataset utilized is described in Section II. The proposed work is covered in Section III. The results are presented in section IV, along with a discussion. The conclusion is found in section V.

II. DATASET DESCRIPTION

Dataset IVb.[8] from BCI Competition III was used, and it was provided by the Intelligent Data Analysis Group and the Department of Neurology, Neurophysics Group (Germany). One healthy person provided the data for this study. The experiment included two motor imaging sessions for the left hand and right foot. The information from the seven initial sessions without feedback is contained in this data collection. To perform two motor imageries (L) left hand and (F) right foot, visual signs (letter presentation) were displayed for 3.5 seconds. Continuous EEG signals from sessions 4 to 7 were given without any cue information (neither target class nor timing) as a test set, while training sets were given entire marker information showing where the mental task was completed.

Motor Imagery(MI) signals on 118 channels were recorded in accordance with the worldwide 10-20 system. For the competition, there were 210 indicators for the left hand and right foot. The data utilized in the investigation was captured at 100 Hz.[8]

III. THE PROPOSED WORK

In this paper, the pre-processing of the EEG signals is discussed. There are various artifacts like ECG, EMG, and eye movement artifacts present in the EEG due to a mixture of signals coming from different parts of the brain due to volume conduction. This problem is more relevant in brain-computer interfaces (BCIs) focused on motor imagery, since the sensorimotor rhythm (SMR) is low and there are some strong activities in the same frequency range (7- 30 Hz), such as visual activity and movement artifacts. Three pre-processing methods are described in the following sections to reduce the various artifacts present in EEG signal.

A. Band-pass Filtering

According to the given dataset, EEG signals are sampled at 1000 Hz since the EEG signal is a low-frequency signal, they are down sampled to 100 Hz for further processing. EEG channels are retransmitted via a 7 to 30-Hz bandpass filter, as mu (μ) and beta (β) patterns are in this range of frequencies.[2].

B. Common Average Referencing(CAR)

To reduce the various objects, present in the high dimensional EEG data, several filtering techniques have been suggested. A common average referencing (CAR) approach is one that reduces the unnecessary EEG components. The average of all electrodes is subtracted from each electrode value in the CAR.

$$e_i^{CAR} = e_i - \frac{1}{N} \sum_{j=1}^N e_j \quad (1)$$

Where N is the number of electrodes.

The CAR eliminates the activities present in large areas of the scalp and can be known as a high-pass filter. Therefore, this improves local activities such as mu and beta activity used in motor imaging. The CAR generates an almost reference-free signal and decreases the noise and artifacts provided by many electrodes, such as artifacts of the visual alpha pulse, Electrocardiogram, Electromyogram, and eye movement, thereby improving the SNR. The effect of applying CAR for different frontal and motor electrodes is shown in the result section.

C. Surface Laplacian

The EEG scalp potential of the surface Laplacian is generally measured in BCI investigations based on two primary assumptions. The electrodes are equidistant, and the surface of the scalp in the near-electrode region is essentially flat. The estimate suggested by Hjorth [6] is

$$e_i^{LAP} = e_i - \frac{1}{4} \sum_{j \in S_i} e_j \quad (2)$$

where e_i is the voltage recorded by the i^{th} electrode and S_i is the index set of the four surrounding electrodes. S_i contains

the nearest-neighbour electrodes in small Laplacian and the next-nearest-neighbour electrodes in large Laplacian.[3]. In Fig. 1, the nearest neighbour electrodes in small Laplacian are represented by a circle and next-nearest neighbour electrodes in large Laplacian by a rectangle.

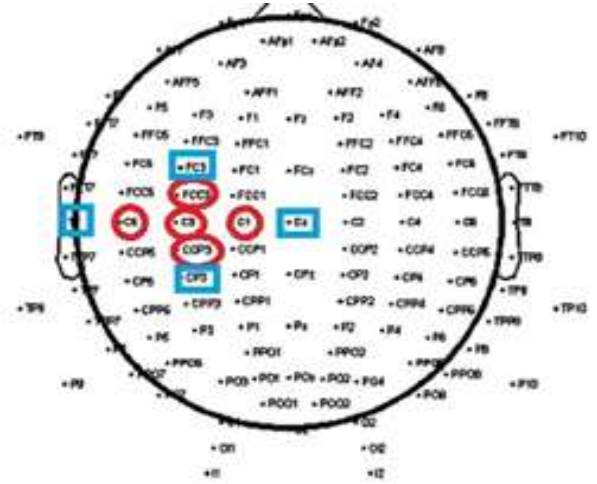


Fig. 1. Nearest electrodes of C3 for the small and large Laplacian in an Extended International 10-20 system.[7]

IV. RESULT AND DISCUSSIONS

Electrodes attached to the frontal lobes are shown in this section which reflects the eye-blinking artifacts in nearly all electrodes up to F6. In frontal-polar electrodes (Fp1, Fp2, etc), blinking creates a wave with a very large amplitude and with very low frequency(<5Hz). A total of 7 electrodes are presented in Fig.2. a) representing eye blinking between 700 to 800 samples having maximum -

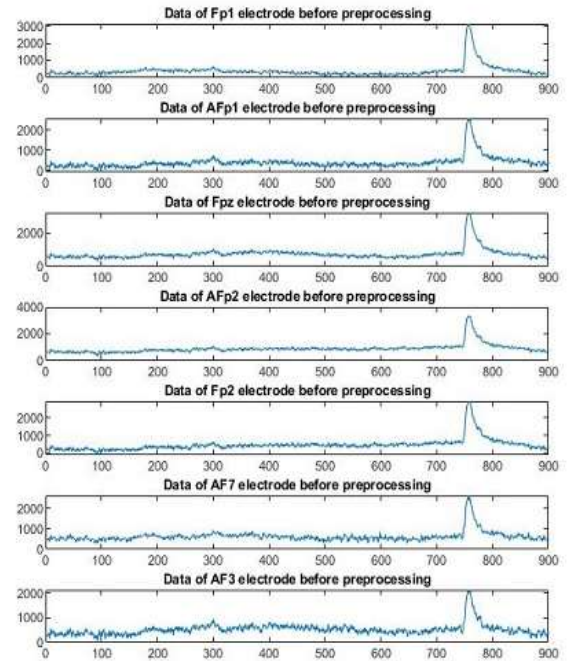


Fig. 2. a) Frontal-polar electrodes (1-7 position) representing eye-blinking artifact.

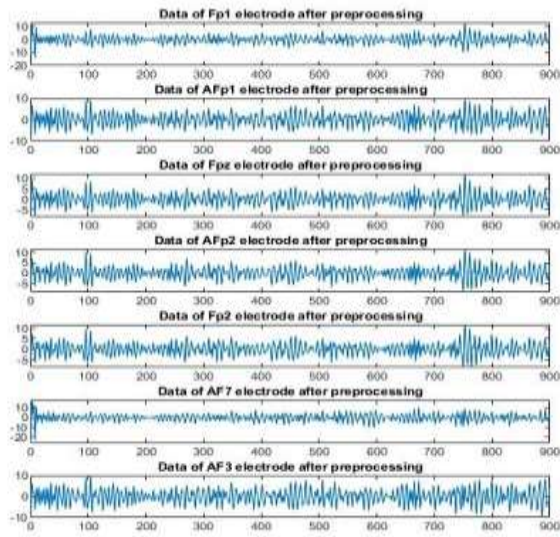


Fig.2.b) Frontal-polar electrodes (1-7position) after pre-processing.

-amplitude at approximately 750th samples. It is visible in almost all electrodes including motor electrodes, but its amplitude decreases as the distance from the frontal electrodes increases. Fig 2. b) shows frontal-polar electrodes from Fp1 to AF3 with CAR pre-processing and effective removal of eye-blink.

Frequency domain representation of the Fp1 electrode representing eyeblink located below 5 Hz is shown in Fig.3 and Fig.4 represents the frequency contents showing removal of eyeblink and characterizes mu (7-16 Hz) and beta(16-30Hz) motor imagery signals after applying CAR.

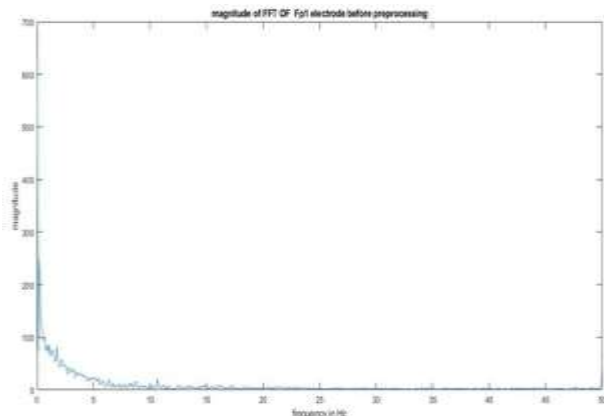


Fig. 3. Frequency contents of Fp1 electrode concentrated below 5Hz indicating eye-blinking artifact.

These signals are related to cortical areas that are linked to the normal motor output pathway of the brain. Action or planning for action results in a decrease in mu and beta patterns. This decrease is called ‘event-related desynchronization’ or ERD. Its contrasting pattern increase, or ‘event-related synchronization’ (ERS) occurs after an action has been executed and with a relaxed condition. The eye-blinking artifact is reduced to a great extent using CAR pre-processing and motor imagination is focused in the range from 7-30 Hz(μ and β rhythms). Hence ERD and ERS which do not need actual action and can happen with motor imagery are most applicable for BCI applications.[2].

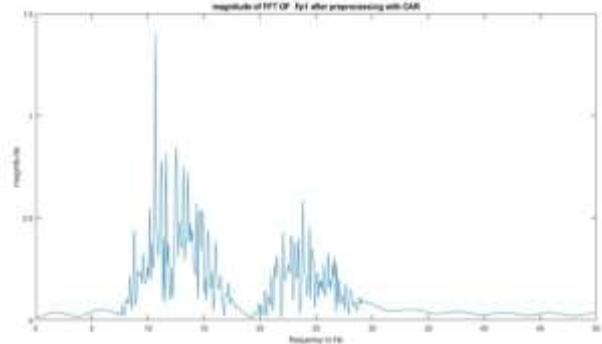


Fig.4. Removal of eye-blinking artifact after CAR Pre-processing on Frontal-polar electrode Fp1

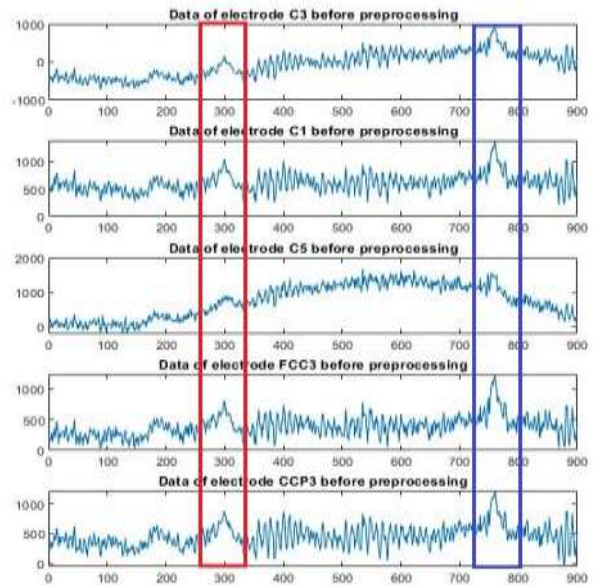


Fig. 5. a) Motor electrodes C3 to CCP3 before pre-processing.

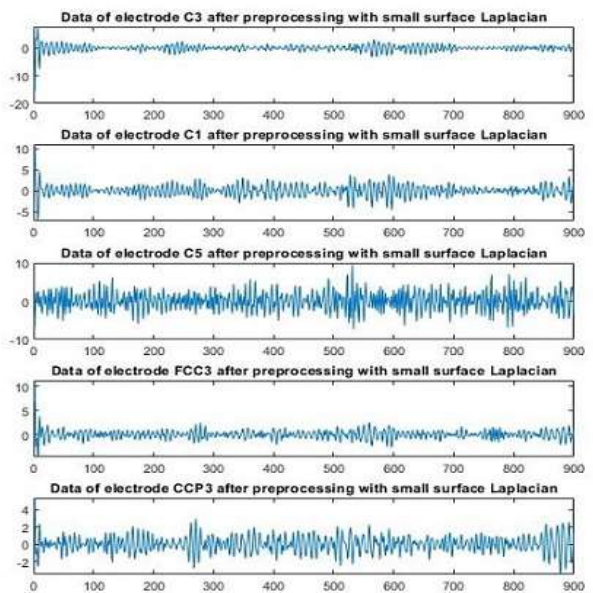


Fig. 5. b) Motor electrodes after pre-processing with Small Laplacian.

Fig. 5 a) and b) represents motor electrodes before and after Small Laplacian. The position of motor electrodes shown by first rectangle in Fig. 5 a) is on motor cortex which is away from frontal electrodes still it is contaminated with eye blinks as shown by second rectangle in Fig. 5 a).

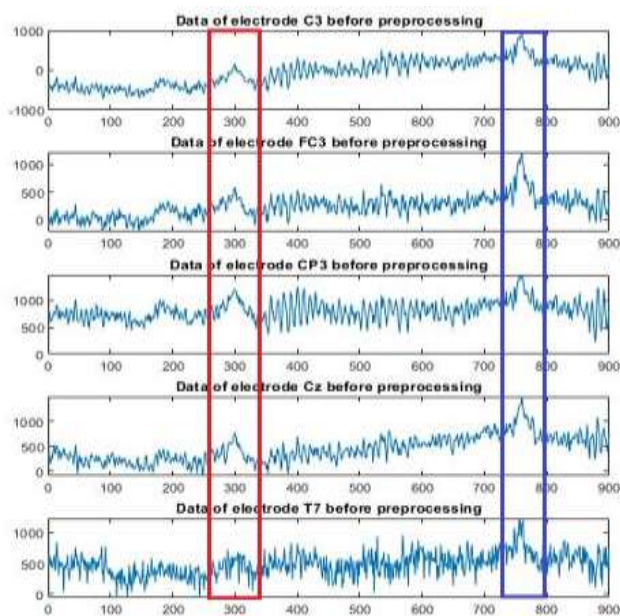


Fig. 6. a) Motor electrodes before large surface Laplacian.

Fig. 6 a) and b) shows the Large Laplacian pre-processing and the reduction of noise and artifacts from the EEG. The electrodes processed in Large Laplacian are shown by rectangles in Fig. 1. Since the surface Laplacian highlights local sources, the number of signals from distant sources will be minimized. The algorithms have been written in MATLAB 2018 in this paper.

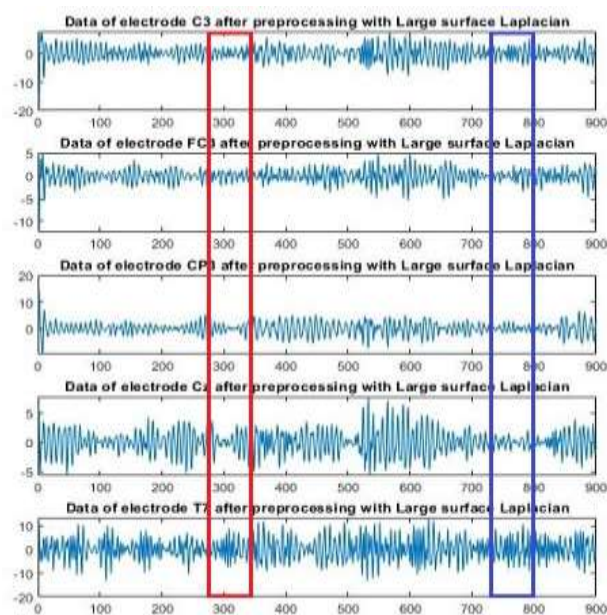


Fig. 6. b) Motor electrodes after large surface Laplacian.

CONCLUSION

In this paper, three pre-processing methods are implemented to reduce the noise and various artifacts received by many electrodes such as ECG, EMG, and eyeblink artifacts. Eye-blinking artifact is reduced to a great extent using band pass filtering and CAR pre-processing and motor imagination is focused in the range from 8-30 Hz (μ and β rhythms). Electromyogram (EMG) artifact is a high frequency signal with frequency range (20Hz -300Hz) that overlaps the EEG signal is eliminated by band pass filtering

(7-30 Hz) and CAR pre-processing. Electrocardiogram (ECG) or pulse artifact is a rhythmic pattern that overlaps the EEG signal. Because the ECG (0-20Hz) frequency components overlap EEG band frequencies, it's impossible to see them with the naked eye.

The CAR eliminates the activities present in large areas of the scalp and improves local activities such as μ and β activity used in motor imaging. The CAR generates an almost reference-free signal and decreases the noise and artifacts provided by many electrodes. The surface Laplacian acts as a spatial filter, lowering spatial noise and limiting the number of available signal sources. The spatial band-pass characteristics of the surface Laplacian are dependent on factors such as the distance between the electrodes and any spatial smoothing applied to the Laplacian. The surface Laplacian generates signals that are less coupled with neighboring channels. The large Laplacian is preferable for estimation, while the small Laplacian is preferable for minimizing the correlation between nearby electrodes.

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